

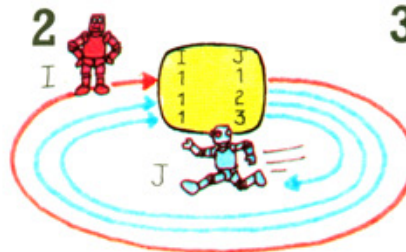
Tricks with loops

Here are some more programs using loops. Below you can find out how you can use loops within loops to repeat several things at the same time. These are called nested loops.

1 Nested loops

```

5 PRINT "I", "J"
10 FOR I=1 TO 3
20   FOR J=1 TO 3
30   PRINT I, J   J loop
40 NEXT J         I loop
50 NEXT I
60 END
  
```



3

I	J
1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

This program has an I loop and a J loop. The J loop is nested inside the I loop and for each time that the I loop is carried out, the J loop is repeated three times, printing

out the value for J each time. The picture above shows the result of this program. The commas spaced the figures out like this.

Computer clock

```

5 CLS
10 LET M=0
20 LET S=0
30 FOR M=0 TO 59
40   FOR S=0 TO 59
50   PRINT M;" ":"S
60   CLS           Seconds
70 NEXT S         loop
80 NEXT M         Minutes
90 END            loop
  
```

0:45

To set the time use this delay loop:
54 FOR Z=1 TO 100
58 NEXT Z

Inside a computer there is an electronic "clock" which sets the rhythm for all the computer's work. The clock pulses at between one and four million pulses a second. This program makes the computer behave like a digital clock. It has nested loops, one to count the seconds and one to count the minutes.

Bugs in loops

```

10 FOR I=1 TO 4
20   FOR J=1 TO 4
30   PRINT I
40   PRINT J
50 NEXT I
60 NEXT J
  
```

Both parts of a nested loop must be inside the other one.

The seconds loop is carried out 59 times for each minute loop. If you try this program on a computer it might run very fast at first. You need to put in an extra "delay loop", then set it by changing the figure in the loop so your computer clock "ticks" at the same rate as a real one.

Random number tester

```

10 FOR I=1 TO 1000
20 LET R=INT(RND(1)*6+1)
30 IF R=1 THEN LET A=A+1
40 IF R=2 THEN LET B=B+1
50 IF R=3 THEN LET C=C+1
60 IF R=4 THEN LET D=D+1
70 IF R=5 THEN LET E=E+1
80 IF R=6 THEN LET F=F+1
90 NEXT I
100 PRINT "FINISHED"
110 PRINT A, B, C
120 PRINT D, E, F
130 END
  
```

This program takes a long, long time. You can make it shorter by changing the number in line 10 to 500 or even 250.

RUN
FINISHED

162	168	167
160	187	156

This program shows if RND really works. The loop from lines 10 to 90 makes the computer pick a random number between 1 and 6 a thousand times. It keeps count of how often each number is picked in the variables A to F, then prints out the results.*

*On some computers, e.g. ZX81 and BBC micro, you need some extra lines at the beginning of the program to set each variable to 0.